

Sentence stress and the procedures of comprehension

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Abstract

This chapter develops a relevance-theoretic model – building on and refining the ideas of Sperber and Wilson (1979, 1995) and Breheny (1998) – of how one aspect of intonation, the position of sentence stress, affects the incremental processing of simple utterances in English. After proposing a notation (adapted from Wedgwood, 2005) for modeling how cognitive effects may be derived from not-yet-fully-decoded utterances, the chapter advances the central tenet is that the position of sentence stress should be seen as marking a particular stage in the hearer's incremental processing, encouraging the investment of effort in deriving cognitive effects at that stage. It sketches how this approach compares against other work within Relevance Theory, including efforts to analyze intonational contours or tones themselves in terms of procedural meaning, and it concludes, *contra* Breheny (1998), that although the position of sentence stress has an impact on the procedures of utterance comprehension, it is *not* best viewed as constituting a type of "procedural meaning" *per se*.

I. Introduction

The intonation of an utterance contributes to a variety of aspects of its interpretation, including information structure, illocutionary force, speaker attitude, etc. Of the pitch accents (salient maxima/minima of fundamental frequency) within a certain domain, which here I assume is the whole utterance, I assume that one, the final pitch accent, is the most prominent: this is the nuclear pitch accent, marked with all capitals as in (1), which following Ladd (1996) I call "sentence stress".ⁱ In English the position of sentence stress is largely free; in (1), for instance, any of the three major constituents – subject NP, verb, object NP – may bear sentence stress:

- (1) a. John broke the VASE.
b. John BROKE the vase.
c. JOHN broke the vase.

The position of sentence stress serves in English as a primary means of indicating information structure (a term which goes back to Haliday, 1967), in that it affects the contextual appropriateness of utterances. This is typically modeled in terms of question-answer pairs and the partitioning of an utterance into focus vs. nonfocus (or new information vs. old information, etc.) It is therefore the position of sentence stress, typically marked using the all-caps notation in (1), that is the aspect of intonation of primary interest to much work on information structure, with much less interest shown in the overall intonational contour / tone sequence or

the meaning contributed thereby. Typically the all-caps notation in (1) is implicitly assumed to indicate a falling tone, marking a declarative utterance presumably with some form of neutral speaker attitude. Lambrecht (1994:344), for instance, explicitly considers intonation "to be distinct from the study of focus prosody". Some authors within Relevance Theory (RT) have likewise shown similar a tendency to conceptually dissociate the two aspects of sentence stress (e.g. Breheny, 1998 concentrates solely on position; House, 2006 seems to consciously advocate such a dissociation – discussed below).

Such a distinction – between efforts to analyze how meaning is affected by the *position* of sentence stress and by the intonational *contour* or *tone* used – is arguably (as anonymous reviewers of an earlier version of this chapter have aptly pointed out) more an accident of the way interdisciplinary work has evolved than a reflection of the way utterances are produced/processed. Clearly these are two closely intertwined aspects of the same phenomenon. One aim of the present chapter is to show one way the information-structure functions of intonation might link up to its other functions, or indeed be motivated by them, within a relevance-theoretic account: if we assume tones/contours carry a kind of *procedural meaning* (see below), the impact of the position of sentence stress boils down to the timing aspect of when those procedural instructions are processed, interacting with general processes of decoding and interpretation in complex ways.

The rest of this chapter is structured as follows: Section 2 is an overview of approaches to sentence stress, Sections 3-4 introduce some notions of Relevance Theory; and Section 5 fleshes out the approach by working through a stress-final example. Section 6 introduces the notion of "focal scale"; Section 7 looks at examples when stress is not final; Section 8 considers degrees of predictability and the shortcomings of the focal scale approach; and Section 9 looks at infelicity judgments caused by misplaced stress. Section 10 compares the model developed herein to various currents of thought pertaining to sentence stress in the RT literature, and finally argues against the claim that the position of sentence stress encodes a kind of procedural meaning. Conclusions are in Section 11.

II. Approaches to sentence stress

Ladd (1996) traces the history of various ways the position of sentence stress has been treated. One tradition – the "normal stress" approach, prominently represented by Chomsky & Halle (1968) – assumes that syntactic structure determines an unmarked position for sentence stress (generally on the most deeply embedded constituent). One problem here is how to properly account for the observed semantic/pragmatic effects of deviations from unmarked position. A second, opposing tradition – the "highlighting" approach, prominently represented by Bolinger (1974) – holds that it is futile to try to read the position of sentence stress off syntactic structure: instead, stress marks the constituent that bears the brunt of the speaker's message. The problem here has long been how to satisfactorily formalize this intuitive notion.

A certain compromise view has emerged between these two historical traditions: the widespread "focus-to-accent" (FTA) approach (see Ladd, 1996), whose tenets underlie a great deal of the current work on information structure. On this view, once a sentence has been partitioned into focus/nonfocus, a move sensitive to contextual factors, the position of sentence stress within the focus is fundamentally a matter of syntactic structure – this making the

mainstream FTA compromise fundamentally closer in spirit to the "normal stress" approach than to the "highlighting" approach (albeit mediated by certain pragmatic factors, in a bow to "highlighting"). The effect of sentence stress here is therefore mediated and indirect – serving as a potentially ambiguous indication of the scope of focus and the presence of +Focus feature(s), which are interpreted at LF – rather than being more directly traceable to the position of sentence stress itself.

Within Relevance Theory, Sperber and Wilson (1995:202-217) sketch out a compromise view of a very different sort, one which I will call dynamic and inferential: "dynamic" in that it tries to model the online construction of meaning as left-to-right parsing proceeds, and "inferential" in that it assumes inference is already applied to resolving various ambiguities and anticipating the content of the overall utterance even as successive constituents are being recognized and interpreted. They take the radical-pragmatic stance that F-feature marking and focus are entirely dispensable from grammar, that inferential dynamics of interpretation should alone be enough to account for information-structure effects – a challenge to FTA-style work on information structure.

The possibility I wish to explore here – based on the intuitions of Sperber and Wilson (1979, 1995) and in some respects Breheny (1998), but expressed in a more thoroughly dynamic way akin to Wedgwood's (2005) analysis of information structure in Hungarian – is that there is a more direct link between the position of sentence stress in English and its effect: that it simply draws the hearer's attention to a particular stage of processing as being particularly worthy of an investment of effort. This dynamic, inferential approach thus represents a certain converse compromise to the FTA approach, as it views the position of sentence stress as a fundamentally pragmatic rather than syntactic issue (thus in a sense revitalizing the "highlighting" view), albeit mediated by certain structural aspects of incremental parsing (a bow to the "normal stress" view).

III. Relevance Theory and the role of inference

At the heart of RT lies the pivotal notion that utterance comprehension is governed by a cost–benefit tradeoff. According to the Communicative Principle of Relevance (Sperber and Wilson, 1995:158), every utterance carries a guarantee of its own optimal relevance, entitling the hearer to assume that it will be relevant enough (i.e. that it will generate sufficient *positive cognitive effects* – either establishing new assumptions, or strengthening or eliminating existing assumptions) to justify the *processing effort* required to interpret it. One of the main insights of Sperber and Wilson's (1995:202-217) sketch of dynamic and richly inferential processing is that this cost–effect tradeoff also applies in an anticipatory way, at what I will call a *sub-utterance level*: that a not-yet-fully-decoded utterance may already give rise to cognitive effects, and may achieve relevance if those effects justify the effort so far invested. This suggests the possibility that hearers may opt to devote greater effort at an earlier or later stage in processing a given utterance, and that speakers may try to guide those timing strategies.

As Wedgwood (2005:55) points out, this notion of inferences drawn mid-utterance has fundamental consequences for how we view coding vs. inference. Admittedly, one of the ways that Relevance Theory is distinct from Grice's earlier work is that it recognizes that the *explicatures* of an utterance (the enriched proposition expressed) and its *implicatures* (derived

by inference) are constructed by the hearer in parallel and can mutually fine-tune one another, with the possibility of explicit content being adjusted to satisfy the hearer's expectations of relevance (a view encapsulated perhaps most clearly in Sperber & Wilson, 2004:615-17). Nevertheless, I agree with Wedgwood that the dynamic, anticipatory aspect of this mutual fine-tuning as originally laid out in Sperber and Wilson (1995:202-217), with inferences drawn mid-utterance actually affecting how the remainder is parsed and decoded, unfortunately seems to be largely overlooked even in Relevance Theory, with most work instead presenting the mutual adjustment of explicatures and implicatures in a way that is simplified to what may be called the *whole-utterance level*.ⁱⁱ

In specific, Sperber and Wilson (1995) themselves, when they propose an analysis of sentence stress, fail to capitalize fully on their own insight of inferences possibly being drawn on-line during processing – they essentially reiterate their earlier (1979) account, namely that "different stress assignments induce different focal scales" (Sperber and Wilson, 1995:210). It is in this respect alone, appealing to focal scales, that I disagree with and wish to refine their sketch of dynamic, inferential interpretation. Their "focal scale" is defined as a set of entailments of the complete utterance, and as such is formulated on the whole-utterance level. In what follows, I hope to show that there is a certain incongruity here. I attempt to provide some arguments that the impact of the variable position of sentence stress in English must be analyzed on the sub-utterance level, and that if Sperber and Wilson's (1995) analysis of incremental processing is followed through to its conclusion for sentence stress, a more coherent view arises than their "focal scales" or than Breheny's (1998) related notion of "background assumption schema".

Making this argument will require a way to represent how inferences may be drawn on the sub-utterance level, from partially decoded utterances. The next two sections propose a notation for this, drawing heavily on Wedgwood (2005).

IV. Anticipatory hypotheses

On Sperber and Wilson's (1995) approach, at each stage of processing, the hearer posits "anticipatory syntactic hypotheses" about the overall structure of the utterance as well as "anticipatory logical hypotheses"ⁱⁱⁱ about its overall propositional content.

On the syntactic side, I take it as uncontroversial that (2) is the syntactic structure for the full utterance (1), irrespective of sentence stress:

(2) [S [NP John] [VP [V broke] [NP the vase]]]

(1) contains three distinct syntactic constituents (subject, verb, object), and therefore I assume it may be processed by an attentive listener in (at least) three distinct stages.^{iv} (2) can be seen as the goal towards which the parser is working over the three stages represented in (3):

<i>constituents so far encountered</i>	<i>anticipatory syntactic hypothesis</i>
(3) a. "John..."	[S [NP John] [VP]]
b. "John broke..."	[S [NP John] [VP [V broke] [NP]]]
c. "John broke the vase."	[S [NP John] [VP [V broke] [NP the vase]]]

After each constituent is processed in (3), it is assumed that parsing heuristics suggest something like the "anticipatory syntactic hypothesis" shown here on the right: a tentatively chosen, most likely continuation of the tree, filled with placeholders (empty [_{VP}] and [_{NP}]). Probabilistic aspects of parsing presumably play a significant role here: syntactic hypothesis (3)a, for instance, is informed by the fact that NPs encountered at the beginning of a parse in English are likely to be the subject, (3)b by the fact that the verb "broke" is often transitive, etc.

Authors working in the frameworks of Dynamic Syntax (such as Kempson et al., 2001, 2006) and RT (Vačkov, 1997), taking their cue from this same section of Sperber and Wilson (1995:202-217), have analyzed the role played by anticipatory syntactic hypotheses akin to those in (3) in generating information-structure effects, concentrating mainly on the significance of word order. In this chapter I will instead concentrate on sentence stress and its impact on the incremental construction of the other, semantic type of hypotheses which Sperber and Wilson assume that hearers make at each stage ("anticipatory logical hypotheses").

On the semantic side, then, we need a similar way to represent the semantic representation of the full utterance (1) and how it might be built up from individual pieces as processing proceeds. Within RT, the base-level *explicature* (full propositional form enriched with reference and sense assignments) of a sentence like (1) is customarily represented in natural language, as something like "John Smith broke the vase standing in the corner"; which via inferential processes yields *implicatures* (which for (1) might include "That is the source of the noise we just heard" and "John Smith is extraordinarily clumsy"), also represented in natural language. In this chapter, however, I adopt a more formal semantic notation borrowed and adapted from Wedgwood (2005) and Herburger (2000). This notation uses the epsilon operator ϵ (derived from Egli and von Heusinger, 1995) and decomposed Davidsonian event semantics (derived from Parsons, 1990)^v – each of these two aspects will next be briefly explained.

The epsilon operator ϵ precedes a variable and has the effect of selecting a witness of that variable's restrictor set, provided it is non-empty (otherwise assigning an arbitrary value). Hence the expression $\epsilon, x [\text{cat}(x)]$ can be read as an instruction: "assign a value to x such that x is a cat, provided there are any such values, otherwise assign an arbitrary value to x ." As Wedgwood (2005) underscores, an epsilon expression is a referring expression, not an asserted proposition. But if the same restrictor predicate is applied again to the epsilon expression, in this case yielding $\text{cat}'(\epsilon, x [\text{cat}'(x)])$, the result is a truth-conditional assertion we can paraphrase as "whatever is assigned by the epsilon term, it is a cat". By this logical rule, then, $\text{cat}'(\epsilon, x [\text{cat}'(x)])$ is equivalent to $\exists x. \text{cat}'(x)$, or "some cat exists".

Decomposed Davidsonian event representations, in turn, consist of an event variable e restricted by a conjunction of predicates.^{vi} On this approach, the verbal predicate of the event, plus the tense, arguments, and any adjuncts are all represented as separate predicates over events, restricting the event variable e . These individual conjuncts may be written in arbitrary order. The event described in (1) will be represented as the expression $e [\text{Agent}(e, \text{'John'}) \ \& \ \text{Past}(e) \ \& \ \text{break}'(e) \ \& \ \text{Theme}(e, \text{'the-vase'})]$, which may be read as "an event of breaking in the past such that 'John' is the agent of the event and 'the vase' is the theme."^{vii}

Putting this all together, I assume (4) below as the semantic representation (or base-level explicature) of the full utterance (1) – irrespective of sentence stress:

(4) $break'(\epsilon, e [Agent(e, John') \ \& \ Past(e) \ \& \ break'(e) \ \& \ Theme(e, the-vase')])$

The proposition in (4) can be paraphrased (starting first with the inner epsilon operator ϵ) as an instruction "select an event of breaking in the past, provided there is one, such that 'John' is the agent of the breaking and 'the vase' is the theme of the breaking", plus (continuing now with the outer predicate, $break'$) an assertion that "this event was one of breaking". This in turn, by the rule explained above and with certain simplifications, is equivalent to the existential proposition $\exists e. Agent(e, John') \ \& \ Past(e) \ \& \ break'(e) \ \& \ Theme(e, the-vase')$, "there exists an event such that 'John'..." etc. The verb "break" is represented twice in (4): once as an inner predicate defining the type of event, once as an outer predicate serving to cancel out the epsilon operator and yield an existential assertion.

This full semantic representation (4) can be seen as the goal towards which the dynamic and inferential process of semantic interpretation, in parallel to the above syntactic hypothesizing, is working over the same successive stages. The task is to model how the hearer builds up this semantic representation up over time. I will now look in detail at how the step-by-step processing of a single sentence (1)a-c might proceed with three different stress positions and three different contexts, aiming to formalize and flesh out Sperber and Wilson's (1995) approach with the notation above.

V. A dynamic analysis of utterance-final stress

First, consider (1)a with sentence stress in final position, embedded here into the question-answer exchange in (5)-(6):

(5) Q: What was that noise?

(6) A: John broke the VASE.

Upon having processed the initial constituent "John..." in (6), what might the hearer hypothesize about the semantic content of the overall utterance? A referring expression requires a search to identify its intended referent; Sperber and Wilson (1995:204) maintain that the hearer may have good reason to launch a preliminary search immediately, risking an early investment of effort without waiting for the remainder of the utterance to be processed. By reducing the space of possible interpretations that have to be borne in mind as processing proceeds, such an early preliminary assignment (if borne out) may help reduce processing effort down the line, making the overall utterance more relevant. In RT, searches of this and other types are seen as guided by a general-purpose processing heuristic known as the Relevance-Theoretic Comprehension Procedure, a strategy motivated by the Communicative Principle of Relevance:

(7) The Relevance-Theoretic Comprehension Procedure (RTCP)

Follow a path of least effort in computing cognitive effects:

- (a) Consider interpretations (reference assignments, contextual assumptions, implications, etc.) in order of accessibility.
- (b) Stop when your expectations of relevance are satisfied. (Sperber and Wilson, 2002)

In other words, at this stage in our example the hearer considers possible referents in order of accessibility and makes a tentative assignment of *John* as the first, most accessible referent, about which he can expect the speaker's utterance will be relevant in an accessible context. Let us assume here that there is a single, highly accessible candidate (e.g. a certain John Smith is present), for which the hearer concludes the utterance will likely be relevant as a response to (6). Because a speaker aiming at optimal relevance should phrase her utterance so as to facilitate correct, early disambiguation, the Communicative Principle of Relevance entitles the hearer to tentatively assume that his first assignment is correct. (I use the standard convention of "she" for speaker and "he" for hearer).

This tentative reference assignment plus the anticipatory syntactic hypothesis in (4a) (conjecturing that this first-encountered NP is the subject of the utterance) yield an anticipatory logical hypothesis.^{viii} A first attempt at representing it might be (8):

(8) $\epsilon, e [\text{Agent}(e, \text{John}')]$

(8) can be paraphrased as "select an event such that 'John' is the agent of that event".^{ix} However, (8) does not yet exhaust everything the hearer may hypothesize about the overall utterance at this early stage. In Wedgwood's (2005:204) dynamic and inferential account of Hungarian, which I continue to draw upon heavily, the first constituent of any utterance is assumed by the hearer to be part of a full proposition and thus "carries the expectation of the essential elements of a proposition", namely two mandatory kinds of restrictions every event variable must have: a verb-like description of the kind of event and a temporal anchor associated with tense. I assume this carries over straightforwardly to English and here, again, I largely adopt Wedgwood's conventions. Adding two conjuncts reflecting these "essential elements" to (8) yields (9)a below – what I assume is the first anticipatory logical hypothesis, a kind of skeletal proposition:

<i>constituents so far encountered</i>	<i>anticipatory logical hypothesis</i>
(9) a. "John..."	$\epsilon, e [\text{Agent}(e, \text{John}') \ \& \ T(e) \ \& \ V(e)]$
b. "John broke... "	$\text{break}'(\epsilon, e [\text{Agent}(e, \text{John}') \ \& \ \text{Past}(e) \ \& \ \text{break}'(e) \ \& \ \text{Theme}(e, Y)])$
c. "John broke the VASE."	$\text{break}'(\epsilon, e [\text{Agent}(e, \text{John}') \ \& \ \text{Past}(e) \ \& \ \text{break}'(e) \ \& \ \text{Theme}(e, \text{the-vase}')])$

T and *V* here are semantic placeholder-predicates analogous to the syntactic placeholders $[_{NP}]$ and $[_{VP}]$ used above, and may be thought of as *metavariables* in the sense of Kempson et al. (2001:34-35): not variables that range over objects but substitution-sites for the right kind of semantic material. The representation in (9)a can be paraphrased as an instruction: "select an event involving some predicate *V* at some temporal anchor *T* with 'John' as its agent".

One benefit of adopting such an event-structure notation to formalize Sperber and Wilson's account is that it can model in greater detail how the hearer builds up the semantic value of an utterance as it unfolds, including a varying mixture of conjuncts derived from the decoding of already-processed constituents, resolved or disambiguated via inference (in (9)a,

Agent(e, John') is of this kind), as well as conjuncts whose very presence and/or partial disambiguation are derived from inferential conjecture alone (in (9)a, ***T(e)*** and ***V(e)*** are of this latter kind). To represent this more clearly, in this chapter I add to Wedgwood's (2005) notation the additional convention of writing elements of the latter kind – derived from inferential conjecture alone – in boldface.

The representation in (9)a could of course be subject to still further enrichment at this first stage of processing. Sperber and Wilson's (1995:207) account would explain it this way: a tentative reference assignment performed upon processing of the partial utterance "John..." may raise what they call a "relevant question", the answer to which "is certain or likely to be relevant" in a context accessible to the hearer (in our case "What did John Smith do?"). With the notation used here, this same idea can be captured by claiming that merely by being introduced into the representation, the metavariables *V* and *T* may encourage the hearer to risk devoting some effort to hypothesizing about their ultimate values. The logic here is the same as above: again an early search, if successful, may bring the payoff of lower overall processing effort down the line.

However, at this stage in the question-answer context of (5)-(6), there may not be any single, most prominently accessible values to assign to *V* or *T* which the hearer has grounds to expect to be relevant (as we assumed above there was for resolving the referent of "John"). Rather, the hearer's search – assuming that he does risk devoting effort to resolving the metavariables in (9)a at this stage – may instead turn up a broad and diffuse range of predicate/tense assignments possibly pertaining to John Smith, more or less of equally low salience, for which the hearer may expect the overall utterance to be more or less equally relevant (e.g. by somehow explaining the source of the noise). Such a set of possible continuations might be represented loosely as: {*dropped his glass, is smashing things again, knocked over the table, broke the mirror, broke the vase...*}. It would be hard for the hearer to successfully draw any inferences at this stage, abstracting over this diffuse set.

Now, consider how this emerging representation may develop at the next stage of processing, as "broke" is encountered. A finite verb in English may be seen as contributing at least three types of information enabling the hearer to further flesh out the skeletal proposition in (9)a, yielding the representation in (9)b. Firstly, "broke" supplies tense information, thus replacing the metavariable *T* with a specific tense. Secondly, it supplies the verbal predicate involved in the event, thus replacing the metavariable *V* with a concrete predicate – with the caveat that here, again, sense disambiguation is required (which sense of the word "break" is intended). As above this is resolved by the hearer by following the RTCP, i.e. by preliminarily choosing the first, most accessible sense that can be expected to yield adequate cognitive effects. If the sense so chosen, call it *break'*, is transitive, this triggers/reinforces the transitive anticipatory syntactic hypothesis in (3)b and leads the hearer to assume that a posited object NP will play a role in the event as a theme argument – this in turn leads a different placeholder, a variable *y*, to be added to the representation (as a conjunct marked in bold). Thirdly, I assume that it is upon processing a finite verb like "broke" that in English the hearer can infer – at least on purely syntactic, if not yet intonation-contour grounds – something about the illocutionary force of the utterance (that it is likely an assertion, not a question, negation, command, etc.). All of this is modeled as (9)b.

Like before, the mere presence of the variable *y* triggers a search to resolve its value, again following the RTCP – in Sperber and Wilson's terms, this would be seen as (9)b raising the relevant question "What did John break?". Again, assuming the hearer risks devoting effort immediately to resolving *y* (in effect, seeking to answer the relevant question), the RTCP may not turn up any single most accessible assignment likely to be relevant, but rather a range of more or less equally accessible possible values, which he has grounds to expect will be more or less equally relevant. Such a diffuse set of possible continuations, not narrowing down the search space very much, might be represented loosely as: {*a glass, his leg, the mirror, the vase...*}.

Now, consider the final stage of processing (6), when the hearer encounters the final constituent "...the VASE". At this final stage, as Sperber and Wilson (1995:208) put it, if the speaker has succeeded in achieving optimal relevance the referent of the vase "should fall into a place already prepared for it during the interpretation process". In their terms, this means answering any relevant questions raised but left unanswered. In my terms, the value *the-vase*' is simply added to the representation in (9)b, replacing the last remaining variable *y*, to yield the full explicature in (9)c.

So far, what I have presented may be seen as simply a more detailed working-out of Sperber and Wilson's (1995) account of incremental interpretation. However, I want to explore here an analysis of sentence stress that differs from theirs: I claim sentence stress simply draws the hearer's attention to a particular *stage* of processing as being particularly worthy of an investment of processing effort, holding forth the promise of rich cognitive effects to be derived at that stage (I will compare this claim to their focal-scale approach in more detail below). When sentence stress is in final position, as in (6), the hearer is in an advantageous position to derive cognitive effects (e.g. implicatures possibly including "John is clumsy" and "The speaker is not particularly fond of John") having already received, as it were, all the pieces of the puzzle before being invited to do so (i.e. having constructed the full explicature).

Several times in this section I have written that while constructing hypotheses the hearer may "risk" investing a certain effort into preliminarily disambiguating/enriching unresolved elements of those hypotheses – suggesting a kind of gambit that an earlier investment of effort might hold down overall processing effort and result in greater relevance of the overall utterance. In the case of a stress-final utterance, such an investment is "risky" from the hearer's standpoint in the sense that the speaker has not especially encouraged such early investment.

VI. Focal scales

We can now make a first pass at comparing my proposal that sentence stress marks a stage in processing to Sperber and Wilson's approach to sentence stress, which can be traced back to their early (1979) account of ordered entailments. Of the many logical entailments of an utterance, some are more important in its pragmatic interpretation than others; Sperber and Wilson (1979) maintained that a speaker uses sentence stress (which they call "focal stress") to guide the hearer by indicating the "pragmatically most important entailments" of an utterance, "picking out and bringing to the forefront" a subset of those entailments. This subset – called a *focal scale* – is derived from the whole utterance via variable substitution, starting from the

constituent bearing stress and moving up the syntactic tree. The focal scale for (6), for instance, is shown in (10) (with square brackets added for clarity):

- (6) John broke the VASE.
 (10) a. John broke [something]. (replacing the constituent bearing stress, object NP)
 b. John [did something]. (replacing next highest syntactic constituent, VP)
 c. [Something happened]. (replacing whole S)

As Sperber and Wilson (1979) argued, the position of sentence stress indicates that the entailments on this resulting focal scale are to be considered with priority in establishing an utterance's relevance – thus imposing a constraint on the context of interpretation.

In their subsequent book, Sperber and Wilson's (1995:210) analysis of the position of sentence stress, stating that "different stress assignments induce different focal scales" is – despite being presented in the same section that pertains to dynamic processing – essentially a reiteration of their earlier (1979) non-dynamic account.

For the stress-final example above, the focal scale account succeeds just as well as my approach does: each of the entailments in (10) does indeed play a role in the pragmatic interpretation of (6), coinciding precisely with a "relevant question" raised at a certain stage (in inverse order). My account merely suggests in more detail how these questions arise: the presence of any variables, metavariables, or weakly conjectured content in the semantic representation may lead the hearer to devote effort to further resolving them.

VII. A dynamic analysis of non-final stress

Next, consider (1)b, where the speaker chooses to place sentence stress on an earlier, non-final constituent, embedded into the question-answer exchange in (11)-(12):

- (11) Q: Where's the vase?
 (12) A: John BROKE the vase.

On a dynamic, inferential account, processing of (12) might be modeled as follows:

<i>constituents so far encountered</i>	<i>anticipatory logical hypothesis</i>
(13) a. "John..."	$\varepsilon, e [\text{Agent}(e, \text{John}') \ \& \ T(e) \ \& \ V(e)]$
b. "John BROKE..."	$\text{break}'(\varepsilon, e [\text{Agent}(e, \text{John}') \ \& \ \text{Past}(e) \ \& \ \text{break}'(e) \ \& \ \text{Theme}(e, y)])$ <i>further enriched to:</i> $\text{break}'(\varepsilon, e [\text{Agent}(e, \text{John}') \ \& \ \text{Past}(e) \ \& \ \text{break}'(e) \ \& \ \text{Theme}(e, \text{the-vase}')])$
c. "John BROKE the vase."	$\text{break}'(\varepsilon, e [\text{Agent}(e, \text{John}') \ \& \ \text{Past}(e) \ \& \ \text{break}'(e) \ \& \ \text{Theme}(e, \text{the-vase}')])$

Through the first stage, as the table shows, processing proceeds identically as for the previous example. Similarly, the next stage of processing in (13)b, when the finite verb "broke" is encountered, once again contributes in four ways: adding values to replace both metavariables *T* and *V*, an outer predicate reflecting assertion, and a conjunct reflecting a posited object, *y*. As above the presence of this variable *y* triggers a preliminary search for its value, but this time – in the context of this question-answer exchange (11)-(12) – rather than

returning a diffuse set the hearer's search following the RTCP will indeed encounter a suitable candidate, the referent of "the vase" in (12), that is very highly accessible in the context and very likely to be relevant as the assignment for y . This assignment is therefore preliminarily made by the hearer, and this is shown in an additional substage of (13)b – where the conjunct nevertheless remains marked in boldface, reflecting that it is still a preliminary assignment grounded in inferential conjecture (albeit strong conjecture), not decoding.

Another difference to the previous stress-final example is that now sentence stress is encountered at this second stage of processing. I claim this encourages the hearer to go ahead and invest effort in deriving cognitive effects at this earlier, non-final stage. Such a claim of course raises questions about what it means to derive cognitive effects from an incompletely processed utterance; this I hope to illustrate by examples. Here in (12), the hearer plainly does not have to take that much of a plunge to do so: all he has to do is assume that the preliminary assignment he has made for y , following the RTCP, is the one the speaker intended him to make, and that it will be borne out at a subsequent stage (an assumption underlying *all* assignments and disambiguations). Sentence stress – an indication from the speaker that she expects it to be worth the risk to invest effort early (thereby reducing processing costs down the line, thereby increasing relevance) – serves as further encouragement to treat those assignments as correct.

What happens after the constituent bearing sentence stress is processed? In this example, the third and final stage of processing merely confirms this already quite strong hypothesis. This may be viewed in terms of the familiar RT notion of strengthening as one of the types of cognitive effect, albeit applied on a sub-utterance level. This strengthening is reflected in the notation used in (13)c as the final conjunct changing from boldface to regular type (grounded in decoding). Processing of such a constituent, a by-now largely predictable one, may not be relevant in its own right, beyond "acting merely as a reminder, a confirmation of an interpretation which the hearer should have been able to arrive at unaided" (Sperber and Wilson, 1995:211).

Next, briefly consider the last of our first three worked examples: (1)c embedded into the question-answer exchange in (14)-(15), where the speaker chooses to place sentence stress even earlier:

(14) Q: Who broke the vase?

(15) A: JOHN broke the vase.

On a dynamic and inferential account, processing of (15) might proceed as follows:

<i>constituents so far encountered</i>	<i>anticipatory logical hypothesis</i>
(16) a. "JOHN..."	$\epsilon, e [\text{Agent}(e, \text{John}') \ \& \ \textbf{T(e)} \ \& \ \textbf{V(e)}]$ further enriched to: $\textbf{break'}(\epsilon, e [\text{Agent}(e, \text{John}') \ \& \ \textbf{Past(e)} \ \& \ \textbf{break'(e)} \ \& \ \textbf{Theme(e, the-vase')}])$
b. "JOHN broke..."	$\text{break'}(\epsilon, e [\text{Agent}(e, \text{John}') \ \& \ \text{Past(e)} \ \& \ \text{break'(e)} \ \& \ \textbf{Theme(e, the-vase')}])$
c. "JOHN broke the vase."	$\text{break'}(\epsilon, e [\text{Agent}(e, \text{John}') \ \& \ \text{Past(e)} \ \& \ \text{break'(e)} \ \& \ \text{Theme(e, the-vase')}])$

Here, things proceed differently from the very first stage. In the previous two examples, the hearer may or may not have opted to risk devoting some effort to resolving the values of the metavariables T and V at such an early stage. Processing of (15) differs in two regards: firstly,

investment in this search is now encouraged by the presence of sentence stress at this initial stage, urging the derivation of cognitive effects from the skeletal proposition alone. Secondly, in this minimal question-answer context this search can be assumed to turn up a single most accessible value for both *T* and *V* (the latter being a transitive sense of *break*'), as well as for the object posited on the basis of this transitive sense – all of these assignments very likely to yield a relevant utterance. Moreover, the question-answer exchange gives the hearer at least some grounds to presume the response is likely an assertion. All of this is reflected in the enriched proposition in the additional substage of (16)a. Then, as these hypotheses are successively confirmed by decoded material in subsequent stages, (16)b and (16)c, the corresponding elements change from boldface to normal type.

After these three worked examples, the picture that – I hope – is starting to emerge is that on this approach what the position of sentence stress contributes or communicates in simple English sentences like (1) has a lot to do with processing strategy and the timing of effort. Sentence-final stress suggests something of a default processing strategy to the hearer and does not especially encourage early investment of effort, whereas earlier (non-final) stress encourages the hearer to invest processing effort earlier. Because of this ostensive encouragement to invest effort earlier, by the Communicative Principle of Relevance the hearer is entitled to expect such early effort to hold down the processing cost of the overall utterance and thus boost its relevance.

In Sections 4 and 6, I have presented a notation for modeling how cognitive effects may be derived from not-yet-fully-decoded utterances, i.e. from an anticipatory logical hypothesis that contains a mixture of information derived from decoding and/or from inference. The use of this notation, including the boldface/regular type convention, aims to illustrate clearly how different stages of processing may contribute to relevance in a range of ways: by contributing wholly new conjuncts to the semantic representation, by filling previous placeholders, by resolving ambiguities of various degrees of diffuseness, or possibly by doing little more than confirming already strongly conjectured hypotheses. This notation thus contributes to the development of a model of how the well-known information-structure effects of sentence stress might emerge dynamically.^x In what follows I will use it to try to show how in certain cases my approach to sentence stress fares better than Sperber and Wilson's "focal scales", as well as an elaboration thereof proposed by Breheny (1998).

VIII. Predictability

For the two non-final stress cases we have now looked at, (12) and (15), Sperber and Wilson's (1995) focal-scale account of sentence stress proceeds somewhat differently than for the stress-final example above. The first entailment on their focal scale (which, remember, is derived by variable substitution of the constituent bearing sentence stress itself) for (12) is "John [did something to] the vase", for (15) it is "[Someone did something to] the vase". If these entailments are to play any role in the pragmatic processing of the respective utterances at the stage sentence stress is reached, Sperber and Wilson (1995:) argue, the hearer would simply "have to know how the sentence will end" – otherwise this sentence stress position will be infelicitous.

Breheny (1996, 1998) formalizes this notion more explicitly: he maintains that this first entailment on the focal scale, which he calls the *background assumption schema*, is constructed in full form at the stage of processing when sentence stress is reached. Just by being so constructed, Breheny argues, such a schema forces the hearer to consider possible alternative values. This demands processing effort, and by the Communicative Principle of Relevance the hearer is entitled to expect that effort not to be gratuitous. Breheny thus concludes that the use of sentence stress, like pointing at something physically, *forces* certain assumptions to play a part in interpretation – he calls this the "pro-active" (Breheny, 1996) or "procedural" (Breheny, 1998) effect of sentence stress. This leads him to propose a generalization about *predictability*: because any and all constituents following the one bearing sentence stress must be represented in the background assumption schema, they generally must be *predictable* (i.e. accessible or introducible via bridging inferences based on encyclopedic knowledge) *before* the stage of processing when sentence stress is reached – otherwise, again, this position for sentence stress will be infelicitous.

Both Sperber and Wilson's focal scale and Breheny's generalization work well enough for the two examples of non-final stress analyzed in the previous section. In both (12) and (15), the hearer following the RTCP found sufficiently salient contextual assumptions to be able to enrich the partial representation and arrive at a full explicature filled out with at least preliminary values (still shown in boldface) at the point when sentence stress is reached; it is in this sense that the hearer knew – or could guess – how the sentence would end. Subsequent stages merely served to confirm those choices. However, not all post-stress constituents are predictable in this straightforward way, and such cases cannot be correctly accounted for by either Sperber and Wilson's focal scale or Breheny's generalization. Firstly, consider another possible answer to the question from (11) above, repeated here:

(11) Q: What happened to the vase?

(17) A: John **BROKE** the darned thing.

At the stage in (17) when sentence stress is encountered, "the darned thing" is hardly predictable material (to the extent it is not a repetition of or derivable from any explicit previous material), although its referent is just as predictable as the referent of the object in (12). Both Sperber and Wilson's focal scale account and Breheny's generalization fail in the same way here, precisely because each requires a specific entailment of the *whole* utterance, namely "John [did something to] the darned thing" (Sperber and Wilson's first entailment on the focal scale, Breheny's background assumption schema), to play a role in processing at the stage when sentence stress is reached.

To capture this point, Sperber and Wilson's and Breheny's account of sentence stress would at least have to be rephrased more abstractly in terms of the explicature of an utterance, not the utterance itself. On my account, such examples can be handled quite naturally. The anticipatory logical hypothesis for (17) at the second stage of processing will be identical to the same stage for "John **BROKE** the vase" in (12): the same enriched anticipatory logical hypothesis shown in (13)b, with *the-vase* as a conjectured (boldface) value inserted for *y*. Sentence stress on "broke" in both (12) and (17) – rather than triggering two different focal scales – simply encourages the hearer to invest effort in deriving cognitive effects at the same stage. The

cognitive effects derivable at that stage, what we might describe as the *main implicatures* of the utterance (such as "John's breaking the vase is the source of the noise"), are those for which the choice of referring expression to follow is immaterial – the object NP could take the form of "the vase", "it", "the darned thing", etc. What we might call *secondary implicatures* (here for instance possibly involving the speaker's attitude towards the vase) may be derived later from post-stress material. (17) thus illustrates one way post-stress material may achieve relevance in its own right, by doing more than just confirming an already-made preliminary assignment or matching up with a focal-scale entailment / background assumption schema (see for instance Scott (2008) on the "extra work that referring expressions do" beyond just referring).

A more serious problem for Sperber and Wilson's focal scale account and Breheny's generalization – again, because they are posited on the whole-utterance level – is that they suggest predictability is an all-or-nothing distinction, whereas not all post-stress material is predictable to the same degree. A post-stress constituent may be rendered partly, but not entirely predictable by a previous constituent, as in (19), with one crucial stage in processing illustrated in (20):

(18) Q: "How did you know I was in town?"

(19) A: "I saw your MOTHER yesterday."

<i>constituents so far encountered</i>	<i>anticipatory logical hypothesis</i>
(20) "I saw your MOTHER..."	$see'(\epsilon, e [Agent(e, I') \ \& \ Past(e) \ \& \ see'(e) \ \& \ Theme(e, your-mother')])$.

Here both approaches would require the entailment "I saw [someone] yesterday" to play a part in processing, requiring "yesterday" to be fully predictable at the stage sentence stress is reached – otherwise sentence stress in this position should be infelicitous. Clearly this is incorrect. Yet on the other hand, "yesterday" is not completely unpredictable, either.

On a dynamic, inferential account, at the stage when sentence stress is processed in (19), the hearer will have derived something like the explicature in (20), with every element derived from decoding rather than inference alone, clearly offering a suitable basis for deriving inferences. The main implicatures of (19) (possibly including "Your mother told me") are again derivable at this stage when sentence stress is reached, and again abstract away from the contribution of any post-stress constituents (for instance, the main implicatures would be the same if instead the utterance had ended "...last weekend" or had simply stopped at "MOTHER"). The post-stress constituent "yesterday" achieves relevance merely by narrowing down part of the meaning contributed by the conjunct *Past(e)*, already added to the representation upon processing of the verb "saw".

A third problem is that the predictability of post-stress constituents cannot always be traced to material processed prior to the constituent bearing sentence stress, but in fact may arise from the constituent bearing sentence stress itself. Moreover, especially in such cases, there may be more than one continuation that is more or less equally predictable, and can be expected to be more or less equally relevant. Classic examples include (22), with an attempt at illustrating how processing might proceed in (23):

- (21) Q: Why are you late?
 (22) A: My CAR broke down.

<i>constituents so far encountered</i>	<i>anticipatory logical hypothesis</i>
(23) a. "My CAR..."	$\epsilon, e [Theme(e, my-car') \ \& \ T(e) \ \& \ V(e)]$ <i>further enriched to:</i> $\left\{ \begin{array}{c} \textit{break-down'} \\ \sim \textit{start'} \\ \textit{steal'} \\ \dots \end{array} \right\} (\epsilon, e [Theme(e, my-car') \ \& \ Past(e) \ \& \ \left\{ \begin{array}{c} \textit{break-down'} \\ \textit{start'} \\ \textit{steal'} \\ \dots \end{array} \right\} (e)])$
b. "My CAR broke down..."	$break-down'(\epsilon, e [Theme(e, my-car') \ \& \ Past(e) \ \& \ break-down'(e)])$

Sperber and Wilson's focal scale account and Breheny's generalization, requiring the entailment "[something] broke down" to play a role when sentence stress is reached, would have to see "broke down" as being fully predictable at that stage. But intuitively, like for the previous example, the post-stress material is here not fully predictable (not being a repetition of or derivable from any explicit previous material), yet is not entirely unpredictable either. Sperber and Wilson (1995:211-212) themselves suggest an alternate account of thetic or all-new sentences like (22), further explored in Sax (2010, in press), where the post-stress material "broke down" merely instantiates one of an array of ways a car may provide a plausible excuse because of what the hearer knows about cars. Its partial or weak predictability thus arises from encyclopedic knowledge triggered by the stressed item itself, at the stage when it is processed – violating Breheny's generalization.

In my terms, by using early stress in (22) the speaker encourages the hearer to already derive what I have called the main implicatures (e.g. possibly including "the speaker considers her tardiness justified by a car-related excuse"), for which the precise post-stress material is immaterial). This means deriving them from an incompletely resolved explicature containing at least several alternative values for *V*, all more or less equally accessible and potentially relevant – something akin to the set {break down, fail to start, was stolen...}. This set of alternatives is less diffuse than those considered above, and because the position of sentence stress here urges inferencing over it, I have made an attempt at representing the set of alternatives within braces in (23)a: with one pair of braces representing alternative inner predicates defining the type of event, another set of braces representing alternative outer predicates yielding an assertion/negation, both in boldface to reflect their grounding in conjecture, and with members of the two sets of alternatives matching up pairwise. This case illustrates how implicatures may fine-tune explicatures on the sub-utterance level, even as the latter are being decoded – or as Sperber and Wilson (2004:615) put it, how "more specific expectations about how the utterance is intended to be relevant", i.e. as an excuse, may "contribute, via backwards inference, to the identification of explicatures" or at least constrain the alternatives being entertained. Next, the post-stress material achieves relevance by merely picking out and confirming one member of that set (shown in (23)b in regular typeface), giving rise to what I have called secondary implicatures (e.g. "the specific car-related excuse involves breaking down").

To sum up this section, I hope to have outlined how both Sperber and Wilson's focal scales and Breheny's predictability fail to fully explain the effect of sentence stress position in certain examples, precisely because both are formulated on the whole-utterance level, as well as how

such examples can be better accounted for on a fully dynamic, inferential account posited on the sub-utterance level. The latter is better at handling: referring expressions that are not predictable although their referents are, weaker/stronger degrees of predictability, and predictability that arises from the constituent bearing sentence stress (as inthetic sentences).

I have appealed to but avoided defining the intuitive notions of "main" and "secondary" implicatures. A more rigorous treatment is beyond the scope of this chapter, but they might be explained in general terms as follows: both types of implicatures are relevant in their own right, whereas "main" implicatures are to some significant degree more relevant than "secondary" implicatures – cf. some similarities to Wedgwood's (2005:191) notion of "main predication" for the stressed focus position in Hungarian.

IX. Misplaced stress and gratuitous effort

What if sentence stress comes too early or late? How do impressions of infelicity then arise? Breheny (1996, 1998) considers infelicity judgments from the perspective of gratuitous effort, but solely in terms of his background assumption schema, and thus only on the whole-utterance level. The account I am proposing can offer a more nuanced explanation of infelicity judgments – in terms of gratuitous effort on both the sub-utterance and whole-utterance levels. This I will illustrate with a few examples.

If in response to the first context question considered above, (5) "What was that noise?" the speaker had opted for an earlier stress position, such as "John BROKE the vase", this would encourage the hearer to invest effort at what we have called the second stage of processing. However, at that stage the hearer's search will presumably not be successful in finding any possibly relevant value for the posited object *y*. Hence, he may at that point invest further effort in backtracking, reconsidering previous choices such as the transitive sense *broke*', hypothesizing an alternative, intransitive sense, and testing that hypothesis for relevance. But even this extra effort (on the sub-utterance level) may leave the hearer without a workable hypothesis about the full utterance, leaving him unable to derive relevant cognitive effects at this stage – as the use of sentence stress has suggested he should be able to. Thus the hearer has no option but to wait until he has processed the whole utterance, and then, as Sperber and Wilson (1995:208) put it, "try to find out under what mistaken image of him the speaker could have thought that her utterance would be optimally relevant". This extra, presumably gratuitous effort on the whole-utterance level may involve the hearer trying to sort out why the speaker thought he should have been able to anticipate at that earlier stage what the object of the breaking event would be (in effect, accommodating a presupposition).

If sentence stress falls still earlier, i.e. if in response to the first context question (5) "What was that noise?" the speaker had opted for an even earlier stress position, "JOHN broke the vase", the hearer would be put to similar but possibly greater gratuitous effort. First on the sub-utterance level, as decoding is still underway, this stress position encourages what will presumably be a fruitless search for salient and potentially relevant values of *T* and *V* already at the first stage of processing. Next on the whole-utterance level, after the whole utterance is processed, it may force the hearer to consider why the speaker thought that some event of vase-breaking should have been accessible and relevant to him at that early stage – here, too, the hearer might backtrack and reconsider which previous aspects of the exchange he may have

misheard/misinterpreted, consider whether the speaker purposefully intended a rhetorical effect, etc.

Conversely, if stress comes too late – e.g. if in response to the third context question considered above, (11) "Where's the vase?" the speaker had instead opted for a later stress position, responding "John broke the VASE" – the result will again be extra, possibly gratuitous effort. At the second stage of processing for this question-answer pair, the hearer should already be in a position to anticipate with some certainty the theme of the breaking event, but the absence of sentence stress at that stage suggests that the speaker does not expect him to be. On the sub-utterance level this may trigger extra effort spent on a further search for other possibly relevant values for *y*, and/or backtracking to consider senses other than *break*, etc. On the whole-utterance level, this may subsequently lead the hearer to devote extra effort to entertaining various contrastive/rhetorical interpretations (such as "...but luckily he did not break anything else") – which may or may not have been intended by the speaker.

X. The "meaning" of sentence stress

The dynamic, inferential approach to the position of sentence stress in English I have advocated here (being a refinement of Sperber and Wilson's (1995) account of incremental processing) might be viewed as a certain revival of the "highlighting" view mentioned in Section 2, grounded in the broader cognitive theory of communication that Relevance Theory provides – here sentence stress marks not so much a particular *word* but a *stage* in processing. In that sense, this approach may be seen as a reassertion of Bolinger's (1983) claim that intonation in English is inherently iconic, in that it posits a direct link between the position of sentence stress and its effect: sentence stress draws the hearer's attention to a particular stage of processing as being particularly worthy of an investment of effort. In what remains of this chapter, I wish to address two related questions: one is *why* the position of sentence stress should have such an effect, and the other is *what type of meaning* – if any – the position of sentence stress should be seen as contributing to the process of interpretation.

Relevance Theory, as is evident from the above discussion, recognizes that linguistic communication involves a complex interaction of decoding and inference, and the chapters in this volume are linked by a concern for the relevance-theoretic distinction between *conceptual meaning* (encoding concepts that figure in semantic representations) and *procedural meaning* (encoding constraints guiding the inferential manipulation of those semantic representations) (see Wilson and Sperber, 1993; Blakemore, 2002; and the other chapters in this volume). As House (2006:1548) puts it, many aspects of prosody simply "cry out for a procedural interpretation" and there is a growing body of RT work in this direction. One main insight in this work is that different contours / tones can be seen as encoding procedural instructions that help to constrain or guide inferential processes of utterance interpretation (for English intonation see Imai, 1998; House, 2006; Wilson and Wharton 2006, Clark, 2007; and references therein; for Spanish intonation see Escandel-Vidal, 1998; Labastía, this volume). For instance, certain English tones have been analyzed as encoding specific constraints on the derivation of higher-order explicatures (i.e. how the proposition expressed is to be entertained, cf. Clark, 2007), or as marking the phrase it accompanies as complete and ready for interpretation or incomplete and to be interpreted as "part of a larger piece of structure" (House, 2006:1554).

I will assume, then, that the contour / tone of the nuclear pitch accent in English encodes such a procedural instruction, guiding aspects of the hearer's inferential comprehension procedures. The question then arises of how the effect of different positions of the nuclear pitch accent should be properly modeled. Overall, at least five interrelated lines of reasoning have been explored within RT in this regard. I will briefly try to explain how my approach differs from each of them and point out some advantages of my approach.

One of them is Sperber and Wilson's (1979, 1995) argument that different stress positions invoke different focal scales, which can be seen as a constraint on the context of interpretation. In the preceding sections, I have tried to show that a fully dynamic approach to sentence stress handles observed facts about post-stress constituents better than focal scales. As I have indicated above, Breheny's (1996, 1998) generalization of predictability is essentially based on the same argument (that the first entailment on the focal scale obligatorily plays a part at the stage sentence stress is reached), and thus suffers from the same problems as the focal scale approach.

A second line of thought (evident in Wilson and Wharton, 2006; also touched upon in Sperber and Wilson, 1995; Breheny 1998) is that different stress positions might achieve different effects simply by requiring lesser or greater amounts of effort, in the latter case raising an expectation for extra cognitive effects. However, both Wilson and Wharton (2006) and Sperber and Wilson (1995) leave it unexplained why specific stress positions should inherently require different effort, seemingly appealing to an unclarified notion of markedness (i.e. why should earlier stress positions inherently be more costly to process – is it purely on frequency-based grounds?). Breheny's arguments about extra effort, in turn, are posited solely on the whole-utterance level – with the same disadvantages as listed above. My approach, by capturing aspects of the cost–benefit tradeoff on the sub-utterance as well as whole-utterance level, as illustrated in Section 9, offers a way to capture more nuanced motivations of markedness and explanations of why misplaced stress is infelicitous.

A third line of reasoning to be found in the RT literature (and elsewhere) is that sentence stress serves as an input to some independent structure-building mechanism, involving dedicated information-structure categories. For instance, one major way in which Breheny's analysis of sentence stress (for which he uses the term "focus") differs from Sperber and Wilson's (1995) account is that he does not adopt the strongest form of their theory, which rejects F-marking and a focus/nonfocus partition as playing any role in the grammar of language. Instead, Breheny assumes – in something closer to an FTA-style analysis – that the constituent bearing sentence stress is marked with an F-feature that gets interpreted at LF, i.e. at the whole-utterance level (on the other hand Breheny's F-feature does not spread or percolate and it marks only a single constituent – this making his use of the term "focus" closely analogous to the way I use "the constituent bearing sentence stress"). Some other authors working on intonation in RT (e.g. Fretheim, 2002; House, 2006:1550) take a similar stance to Breheny's, in the sense that they see the position of pitch accents as contributing to utterance interpretation by possibly manipulating or establishing separate categories such as focus and topic. Herein I have sought to pursue Sperber and Wilson (1995)'s original more radical-pragmatic line – rejecting any such fixed partition or focus structure – for English, following a path largely blazed by Wedgwood (2005) for Hungarian, and sketching out how (at least some)

information-structure effects in English may be generated without appealing to special grammatical mechanisms of information structure or F-marking.

RT has time and again been shown to be a powerful enough pragmatic theory to be able to account for many facts of utterance interpretation while assuming relatively lean and parsimonious grammatical representations (cf. e.g. Scott, 2008 on referring expressions). If the kind of approach advocated here, in Sperber and Wilson (1995:202-217), and in Wedgwood (2005) – eschewing focus as a grammatical primitive altogether – could be sustained for more complex information-structure phenomena in English and cross-linguistically, it would recommend itself at the very least on the grounds of simplicity.

A fourth, again interrelated line of reasoning on sentence stress in the RT literature (and not only) suggests that these information-structure-building properties of the position of sentence stress can essentially be treated independently of the procedural meaning carried by the tone / contour of intonation. For instance, this seems to be stated explicitly by House (2006:1552): "The structure [the speaker] builds directs the listener to the relevant contexts independently of the particular pitch accents used. An understanding of this is essentially separate from the interpretation of the individual pitch accents themselves, or the pitch contour as a whole, which will tell us about the status of the salient material." Rather than seeing these two aspects as somewhat arbitrarily bound together, my account opens up the possibility of viewing the effect of the position of sentence stress as actually being motivated by the interpretation of the pitch accents/contour.

Namely, let us again assume that the different pitch accents or tones do indeed encode different procedural instructions. In the examples used in this chapter, the constituents in all-caps have so far been implicitly assumed (as in much of the information-structure literature) to bear a falling tone. Drawing on Imai (1998), Clark (2007:83) argues that FALL is a "default" tone, bearing a default procedural instruction guiding the construction of higher-level explicatures: "The proposition expressed is entertained as either a description of a state of affairs or as an interpretation of a thought of someone other than the speaker at the time of utterance." The widely noted sense that a FALL tone signals the completeness of an utterance can be seen as resulting from this instruction (cf. the procedural meaning of Clark's LEVEL tone: "This utterance is not yet complete").

Next, let us assume that every instance of sentence stress is the bearer of some such instruction, encoded in its tone, guiding the hearer towards the proper construal of the proposition expressed. It would be natural for a hearer to hold off on devoting effort to deriving cognitive effects from a partially-decoded utterance until that instruction – bringing what is presumably a crucial contribution to utterance interpretation and the construction of at least higher-level explicatures – is processed. This, in turn, gives the speaker of a language with free stress position like English the ability to encourage the hearer to hold down early effort by delaying the delivery of that instruction, until such time as he expects the hearer is in a position to derive what I have called the main implicatures of the utterance. Conversely, the speaker may instead opt to bring that instruction forward in time, encouraging early effort on the hearer's part. When viewed from the dynamic, inferential perspective developed herein, therefore, the position of sentence stress in English essentially becomes a question of timing (relative timing between the incremental processing of the linear sequence of lexical constituents and processing of the procedural instruction carried by the supersegmental pitch

accent). Information-structure effects of sentence stress position are here not just arbitrarily associated with the nuclear pitch accent – rather, it is precisely the procedural instruction borne by the nuclear pitch accent that triggers them.

The fifth and final line of reasoning evident in the RT literature to be discussed here is summed up in Breheny's (1998:110-11) claim that the position of sentence stress itself encodes a specific kind of "procedural meaning", one which manipulates the context of interpretation in that it "promotes certain hypotheses over others".^{xi} This claim apparently rests on the assumption that anything that affects the procedures of comprehension must necessarily be taken to encode procedural meaning. To illustrate that this is incorrect, I believe certain parallels can be drawn to the relative ordering of clauses and constituents, and even to the timing aspect of pauses, when these are viewed from a dynamic perspective. First consider (24)-(25):

(24) She left him. He took to the bottle.

(25) He took to the bottle. She left him. (adapted from Carston, 2002)

While different temporal or causal interpretations of (24)-(25) may result from the relative ordering of the two sentences, rather than ascribing a specific kind of encoded meaning to relative order here – truth-conditional, procedural, or otherwise – RT would see the effects as arising out of general inferential principles. The obligatory linearity of language simply necessitates that utterances must be processed in *some* relative order, not simultaneously. Not only clauses but also certain sentence constituents may exhibit freer relative order in English; consider (26)-(27):

(26) On Monday it RAINED.

(27) It RAINED on Monday. (adapted from Sperber and Wilson, 1995)

Here it seems uncontroversial that "on Monday" contributes conceptual, but not procedural meaning. Yet on a dynamic inferential account, the adjunct's position relative to the core clause could be argued to constrain or guide inferential processes – in (26) pre-empting any early effort to derive a value for the temporal anchor *T*, in (27) narrowing down the temporal value already supplied by the finite verb and possibly confirming an earlier hypothesis. On such grounds one might argue, as for instance Vačkov (1997) does, that relative constituent order, when free, encodes a kind of non-truth-conditional, procedural meaning "which constrains implicatures through manipulation of the hearer's choice of context."

Authors like Breheny (1998) and Vačkov (1997) seem to have taken the notion of encoded procedural meaning too far. In short, surely not everything that affects the inferential processes of interpretation necessarily encodes a procedural instruction. As a last case in point, consider (28):

(28) Now I would like to introduce..... the incredible Houdini!

Various interpretive effects may be achieved by the pause in (28) (building surprise, forcing the audience to devote more effort to pondering various performers possibly about to appear),

but it would be incorrect to describe such a pause – not a properly linguistic device – as *encoding meaning* of any type. I believe it is similarly incorrect to see the relative ordering of utterance constituents / whole utterances, or the relative timing of the procedural instruction encoded by sentence stress vs. the individual utterance constituents being processed, as encoding meaning of any type.

Although each of these analogies obviously has its limits, I think they all point in the same direction. I conclude that aspects of relative timing which affect interpretive processes but which do so via general processing mechanisms would be more accurately described as "having an impact on the procedures of comprehension" rather than as "encoding procedural meaning" akin to the procedural meaning that has been ascribed in RT to discourse connectives, pronouns, mood indicators, pitch contours, etc. This view seems at the very least compatible with those of some authors dealing with intonation in RT, for instance with Wilson and Wharton (2006) and Fretheim (2002). Indeed, a quote from Fretheim (2002) may serve to drive this point home: "Some linguistic devices have no conceptual meaning, nor do they encode a specific procedural instruction for the hearer to follow. Rather, they can be said to offer the hearer procedural information by virtue of their interaction with other kinds of linguistic devices in the utterance. A syntactic construction, or a specific linear order of syntactic elements, can provide procedural constraints without actually encoding them." As an anonymous reviewer has aptly pointed out, "procedural" aspects of language, broadly construed, are everywhere – for what are cognitive systems that engage in perceiving, decoding, reasoning but systems of procedures?

XI. Conclusions:

In conclusion, I have made a case for the view that the position of sentence stress in English, if viewed from a dynamic, inferential perspective with possible inferences drawn mid-utterance taken into account, can be seen as simply drawing the hearer's attention to a particular *stage* of processing as being particularly worthy of an investment of processing effort, holding forth the promise of rich cognitive effects to be derived at that stage. On this view, no extra grammatical or pragmatic mechanisms (neither "focal structure" nor "focal scales") are required to explain the observed information-structure effects of sentence stress in English beyond the general principles of Relevance Theory and the assumption that the pitch accent of sentence stress conveys or helps convey a procedural instruction essential in utterance interpretation – thus reiterating and strengthening Sperber and Wilson's (1995:202-217) radical-pragmatic challenge to the standard "focus-to-accent" (FTA) approach to sentence stress. I have presented a notation for analyzing how representations are built up as parsing proceeds, combining contributions from both decoding and inference on a sub-utterance level. I have outlined how this view compares positively against several currents of thought within the RT literature on intonation and sentence stress.

Lastly, I have proposed that the analysis of information structure and prosody in RT would gain if a consistent distinction is maintained between the *procedural meaning* possibly carried by the tone or contour of sentence stress itself, and what I have called the *impact* on the procedures of comprehension (on both the sub-utterance and whole-utterance levels) of how the speaker opts to position the supersegmental nuclear pitch accent (sentence stress) with respect to the linearly-ordered constituents in an utterance. Given that the notion of sentence

stress used herein is a clearly vast oversimplification of much more complex facts of varying degrees of intonational prominence throughout utterances, closer investigation of how timing aspects of these complexities affect and guide the procedures of comprehension on a sub-utterance level should offer a fruitful area of future research.

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- Blakemore, D. 2002. *Relevance and linguistic meaning: The semantics and pragmatics of discourse markers*, CUP
- Bolinger, D., 1983. The inherent iconism of intonation. In J. Haiman (ed.), *Iconicity in Syntax*. Amsterdam: John Benjamins, 97-109
- Breheny, R. 1996. Pro-active focus, *UCL Working Papers in Linguistics* 8
- 1998. Interface economy and focus. In V. Rouchota & A.H. Jucker (eds.), *Current issues in relevance theory*, John Benjamins, 105-139.
- Carston, R., 2000. *Thoughts and utterances*, Blackwell.
- Chomsky, N. and M. Halle, 1968. *The Sound Pattern of English*. Harper & Row, New York.
- Clark, B. 2007. Blazing a trail: moving from natural to linguistic meaning in accounting for the tones of English. In Nilsen, R.A., Appiah Amfo, N.A. and K. Borthen (eds.) *Interpreting utterances*. Novus, Oslo: 79-91
- Egli, U. & von Stechow, K. 1995. The Epsilon Operator and E-Type Pronouns. In: U. Egli et al. (eds.). *Lexical Knowledge in the Organization of Language*. Amsterdam: Benjamins, 121-141.
- Escandell-Vidal, V. 1998. Intonation and procedural encoding: The case of Spanish interrogatives. In V. Rouchota and A. Jucker, (eds.), *Current issues in Relevance Theory*. Amsterdam: John Benjamins. 169-203
- Fretheim, T. 2002. Intonation as a constraint on inferential processing. In: *Proceedings of the Speech Prosody Conference, Aix-en-Provence*, 307-310.
- Haliday, M. (1967). Notes on transitivity and theme in English. Part 2. *Journal of Linguistics*, 3, 199-244.
- Herburger, E., 2000. *What counts: focus and quantification*. MIT Press.
- House, J. 2006. Constructing a context with intonation. *Journal of Pragmatics* 38(10), 1542-1558
- Imai, K. 1998. Intonation and relevance. In R. Carston and S. Uchida. (eds.) *Relevance Theory: Applications and Implications*. John Benjamins, Amsterdam: 69-86
- Kempson, R., W. Meyer-Viol & D. Gabbay, 2001. *Dynamic syntax: the flow of language understanding*, Blackwell.
- Kempson, R., Cann, R., Kiaer, J. 2006. Topic, focus and the dynamics of language. In V. Molnár and S. Winkler (eds.) *The Architecture of Focus*. Berlin, Mouton/de Gruyter: 59-8.

- Labastía, L. this volume. Procedural encoding and tone choice in Buenos Aires Spanish.
- Ladd, D. R., 1996. *Intonational phonology*, CUP.
- Parsons, T. 1990. *Events in the semantics of English: A study in subatomic semantics*. Cambridge, MA: MIT Press.
- Sax, D., 2010. Predictability, non-final focal stress, and word order in English and Polish. In: E. Wałaszewska, M. Kisiełewska-Krysiuk & A. Piskorska (eds.), *In the mind and across minds: A relevance-theoretic perspective on communication and translation*. CUP.
- in press. Not quite 'out of the blue'? Towards a dynamic, relevance-theoretic approach to thematic sentences. In E. Mioduszevska, A. Piskorska (eds.) *Relevance Round Table Meeting II*, Warsaw University Press
- Scott, K. 2008. Reference, procedures and implicitly communicated meaning. *UCL Working Papers in Linguistics* 20, 276-301.
- Sperber, D. & D. Wilson. 1995. *Relevance: communication and cognition*, 2nd ed., Oxford: Blackwell.
- Vačkov, V. 1997. 'Free' word order and relevance: from Prague to London or to London from Prague. *Oxford Working Papers in Philology, Phonetics and Linguistics* 2: 119-135.
- Wedgwood, D. 2005. *Shifting the focus: From static structures to the dynamics of interpretation*, Elsevier.
- Wilson, D. & D. Sperber. 1979. Ordered entailments: an alternative to presuppositional theories. In Choon-Kyu O. & D.A. Dineen (eds.), *Syntax and semantics 11: Presuppositions*. Academic Press, 299-323.
- 1993. Linguistic form and relevance. *Lingua* 90, 1-25.
- 2004. Relevance theory, in Horn, L. and Ward, G. (eds.), *The Handbook of Pragmatics*, Blackwell, Oxford, 607-632.
- Wilson, D. & T. Wharton. 2006. Relevance and prosody. *Journal of Pragmatics* 38, 1559-1579.

ⁱ Pitch accents are of course just one way speakers may cause certain words to be perceived as more prominent (other such factors including loudness, duration, and voice quality). In this chapter I ignore these other correlates of prominence as well as the impact of any secondary pitch accents.

ⁱⁱ At worst, the possibility of mutual adjustment of explicatures and implicatures seems often underappreciated or overlooked in RT, e.g.: "The explicatures of an utterance may then be used to make further inferences, which are the implicatures" (House 2006:1548). At best, such as in Sperber & Wilson's (2004:615-17) summary of the theory and in RT work dealing with lexical pragmatics, the possible mutual adjustment of explicatures and implicatures is expressly recognized but both explicatures and implicatures remain stated on the *whole-utterance level*, as fully-formed assumptions expressed in natural language – without any clear exploration of the possibility that a preliminary decision concerning the contextually appropriate broadening or narrowing of the meaning of "broke" in (1), for instance, may be made before the remainder of the utterance is processed, i.e. at a stage when the emerging explicature of (1) is still something like "John broke..." Instead, the RT lexical pragmatics literature seems to prefer to consider how the meaning of an individual lexeme, such as "broke" in (1), would be broadened or narrowed as part of a fully-formed explicature, here "John broke the vase".

ⁱⁱⁱ "Logical" would perhaps be better recast as "semantic", but I have chosen to stick with Sperber and Wilson's (1995) terminology on this point.

^{iv} The precise nature of what "chunks" may be processed by the hearer is a complex psycholinguistic question; following Sperber and Wilson (1995) I take it as plausible that each sentence constituent – here, each verb, NP argument, or adjunct – warrants its own distinct stage in processing (possibly further divided into sub-stages).

^v The notation I use differs in significant ways from that used by any of these authors, but essentially represents a simplification of Wedgwood's notation – for expository reasons I have chosen to ignore such complexities as the proper representation of theta roles and temporal anchors.

^{vi} Like Wedgwood (2005:195), given limited space I ignore Parsons' (1990) distinction between events and states.

^{vii} The notation *the-vase* is another obvious side-step to save space; in any more realistic semantic representation the definite article – and the possibility of its bearing sentence stress itself – of course require separate treatment. Note also that "theme" is used throughout in its thematic-role sense, not information-structure sense.

^{viii} The order of presentation here might suggest that syntactic hypotheses are necessarily formed prior to semantic ones but in fact – like the interaction between explicatures and implicatures – they are processed in parallel and subject to mutual adjustment. See note ii.

^{ix} Further steps of reasoning are admittedly omitted here, along the lines of "an animate subject is likely to be agent".

^x For instance, the well-known ambiguous scope of focus in the stress-final example (1)(6) may be seen as resulting from the interaction of various contextual assumptions or conjectures made dynamically, affecting the relative relevance of the subsequent stages, whereas the narrow focus readings of (1)(12) and (1)(15) may be seen as resulting from the interaction of appropriately fitting contextual assumption / salient referents with a more or less skeletal proposition at an early stage of its construction, yielding strongly-grounded conjectures that post-stress material merely serves to confirm. Space prevents a more detailed comparison of this approach to information structure to traditional FTA-style, F-marking based approaches, but see Wedgwood (2005) for such an analysis of information structure in Hungarian.

^{xi} This claim may possibly be echoed in Wilson and Wharton (2006:1571): "properly linguistic prosodic signals (e.g. [...] fully grammaticalized aspects of sentence stress and intonation) might be analyzed" in procedural terms.